



SOC Final Fall22 (Solved)

Society, Environment and Engineering Ethics (United International University)



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SOC

Final Exam Fall 2022

1. Quinn is a member of a machine learning research team studying the role of genetic factors in psychological disorders, particularly focusing on how different variants influence social behavior using Machine Learning. To facilitate training model, Quinn built a tool that linked multiple anonymized data sets: an anonymized set of genetic test results accessible only by medical researchers, a publicly available anonymized database of clinical diagnoses, and a custom database of public social networking posts. To preserve anonymity, the tool replaced all personally identifiable information in the social networking posts with quasi-identifiers. Quinn's team was granted approval for a study by their ethics review board (ERB), on the grounds that all data was anonymous and/or public, and all users had opted in to the data collection.

While testing the tool, Quinn discovered a bug that incorrectly linked some records of multiple individuals as a single person. Given that the data sets were all anonymized, the team had accepted that such erroneous matches were likely to occur. The bug increased the expected number of such matches, but only slightly; as such, the bug was classified as low priority. Quinn raised concerns that there may be other such bugs and suggested that the source code be released under an open-source license to facilitate peer review of both the tool and the overall research. However, Quinn's Team Lead, Mr. Heckles denied to do so by mentioning the existence of such bugs to be very common in the prevailing models.

Identify the IEEE and ACM code of ethics that are violated in the above incident and Explain Briefly.

Ans: Quinn is part of a machine learning team studying genetic factors in psychological disorders. They built a tool that links several anonymized data sets to analyze social behavior. The tool anonymizes personal information in social media posts. While testing, Quinn found a bug that caused some individual records to be incorrectly linked as one person. Although this error was deemed low priority by the team lead, Quinn raised concerns about potential bugs and suggested releasing the code for open review. However, the team lead dismissed this suggestion, citing that such bugs are common.

Violated Codes of Ethics:

i) IEEE Code of Ethics (1.3) - "To be honest and realistic in stating claims or estimates based on available data."

- Quinn's team failed to address the potential risks and impact of the bug on the data's integrity, which could mislead research outcomes.

ii) ACM Code of Ethics (1.2) - "Avoid harm."

- By not investigating the bug further or considering the implications of erroneous data linking, the team risks harming individuals whose data may be misrepresented.

iii) ACM Code of Ethics (3.1) - "Ensure that the public good is the primary concern."

- Dismissing the suggestion to open-source the code undermines transparency and public trust in the research process, which should prioritize the well-being of the subjects involved.

Explanation:

The ethical violations arise from a lack of transparency and accountability in handling data integrity issues. By ignoring the potential risks associated with the bug, the team jeopardizes the reliability of their findings, which can impact real-world applications and individuals' lives. Quinn's suggestion for open-source code review would enhance scrutiny and help ensure ethical standards are maintained, promoting greater trust in the research process.

2. Monica recently started a new software analyst job, joining the SQA team. In graduate school, her supervisor had introduced with several members of the team on a number of projects, involving and mentioning Monica's works when he finds necessary. The team had been fascinated by Monica's work and recruited her as she is nearly graduating. Albus, the team's developer lead, had built a name as a brilliant yet impulsive expert in software architecture. His team's contributions were highly mentioning within the domain, with Albus typically claiming main authorship as the lead developer. Their work was also published frequently in popular news media, always with speech only from Albus. Despite the team's repeated achievements, Albus becomes intolerable with strong words for even silly mistakes. He screams at the person and demean them in internal discussion. On some day, female team members have found their names discarded from research manuscript submissions as punishment. Monica soon realizes herself to be the next person as a target when she made a code modification that faces a bit delay the prototype shortly before the main presentation. Albus refused Monica to join the team during presentation. Realizing Albus's attitude was unprofessional and unacceptable. Mr. Lucifer is the manager of the company agreed that the experience was unpleasant to Monica and shameful to the company's reputation.

Identify the IEEE and ACM code of ethics that are violated in the above incident and Explain Briefly.

Ans: Monica has recently joined a software quality assurance (SQA) team as an analyst. She was recruited due to her impressive work during graduate school. However, her team lead, Albus, is known for being brilliant but impulsive, often taking sole credit for the team's work. He frequently demeans team members for minor mistakes and has been particularly harsh towards female colleagues, sometimes even removing their names from research submissions as punishment. When Monica made a minor code modification that delayed a project, Albus excluded her from the presentation, which Monica found unprofessional. Mr. Lucifer, the manager, acknowledged that Albus's behavior was unacceptable and damaging to the company's reputation.

Violated Codes of Ethics:

i) IEEE Code of Ethics (1.5) - "To treat all persons fairly and with respect."

- Albus's behavior towards Monica and other team members, including verbal abuse and exclusion, shows a lack of respect and fairness.

ii) ACM Code of Ethics (1.2) - "Avoid harm."

- Albus's actions create a hostile work environment, which can harm team members' mental well-being and job satisfaction.

iii) ACM Code of Ethics (3.1) - "Ensure that the public good is the primary concern."

- By taking sole credit for team contributions and discarding team members' names, Albus undermines the collaborative nature of the work and the reputation of the organization.

Explanation:

The ethical violations highlight a toxic work environment fostered by Albus's behavior. His actions not only harm individual employees like Monica but also jeopardize team cohesion and the integrity of the company's reputation. Addressing these issues through effective management and fostering a respectful workplace culture is crucial for maintaining ethical standards and supporting employee well-being.

3. Mr. Ortega is a very young garbage-picker. He picks garbage near an industrial area which is currently used for dumping industrial garbage and recently electrical wastes. Ortega started to pick more electrical waste as it pays more for copper wire and other thing. He has gotten a type of rash recently which he thinks is caused by the electric wastes.

One day while picking up trash he heard sound coming from the trash which later exploded. Ortega got hurt from the explosion. Later a news reports that the explosion was caused by some electric waste. X company that uses the area for garbage disposal claimed it wasn't from their waste as a quality control organization has approved their waste to harmless.

How do you consider the action of company X from the engineering ethics perspective? Do you think E waste can be better managed to avoid such accidents? Briefly discuss your points aligning with the concept of Green Technology.

Ans: Mr. Ortega is a young garbage-picker working near an industrial area that dumps garbage, including electrical waste. He started picking more electrical waste for the higher copper prices but developed a rash he believes is due to this waste. One day, while picking trash, he heard a sound that led to an explosion, injuring him. A news report stated the explosion was from electrical waste, but Company X, which disposes of waste in the area, claimed their waste was certified harmless.

Ethical Considerations for Company X:

From an engineering ethics perspective, Company X's actions raise concerns regarding **responsibility and accountability**. Although they claim their waste is safe, the explosion indicates a failure to ensure the safety of their disposal methods. Companies should prioritize the well-being of the community and environment, adhering to ethical standards that prevent harm.

Managing E-Waste:

E-waste can be better managed to avoid accidents through the following strategies:

- i) **Improved Recycling Practices:** Implementing better recycling techniques that safely handle electrical waste can reduce hazards. Companies should partner with certified e-waste recyclers.
- ii) **Public Awareness:** Educating communities about the dangers of improper waste disposal and the importance of recycling can minimize risks associated with e-waste.
- iii) **Green Technology:** Embracing green technologies can lead to safer waste management. For example, developing environmentally friendly materials and better disposal methods can reduce the risks of accidents.

In conclusion, ethical management of e-waste is essential not only for the safety of individuals like Mr. Ortega but also for the broader community. Companies must take proactive measures to ensure their practices do not endanger public safety.

4. Sudo is New social media (a fictional social media) which allow a massive amount of people to post their various day to day activities and pictures. It also allows them to share their locations. Sudo has virtual currency S-bucks. Sudo gives a small amount for S-bucks for daily logins, sharing shopping posts, locations and also for reaching certain amount of likes and shares which addicts people in their loop. Sudo also uses many methods to maximize dopamine loop while using their application. Sudo earns most of its revenue from advertising and selling meta-data of their users without mentioning that explicitly in the Terms and Conditions of Data privacy. It also recently released a press briefing that say how it helps many government organizations by providing information to help keep miscreants at check. It also boasts to have minimum 2 hour online-time for 50% of its user base.

How Ethical do you consider the Revenue Model of Sudo? If you think it as unethical, what should be an ethical policy for Sudo? Discuss with specific logical arguments.

Ans: Sudo is a new fictional social media platform where users can share daily activities, pictures, and locations. It uses a virtual currency called S-bucks, rewarding users for daily logins and engaging with posts. Sudo aims to keep users addicted by maximizing their dopamine responses and earns revenue through advertising and selling user data without clearly stating this in its privacy terms. The platform claims to assist government organizations in monitoring activities and has a significant portion of its users spending at least two hours online.

Ethical Concerns with Sudo's Revenue Model:

- i) **Lack of Transparency:** Sudo does not clearly inform users about how their data is used or sold. This lack of transparency is unethical, as users should know what they are agreeing to.
- ii) **Exploiting User Behavior:** The use of techniques to maximize user engagement through dopamine loops can lead to addictive behaviors, raising ethical concerns about user well-being.
- iii) **Data Privacy Issues:** Selling user metadata without explicit consent breaches ethical standards related to privacy and respect for users' personal information.

Proposed Ethical Policy for Sudo:

- i) Clear Data Usage Policies:** Sudo should explicitly state how user data is collected, used, and shared in easy-to-understand language within its terms and conditions.
- ii) Informed Consent:** Users should actively consent to their data being used for marketing and other purposes, rather than relying on vague statements in lengthy agreements.
- iii) User Well-being Focus:** Sudo should implement features that promote healthy usage patterns, such as reminders to take breaks and tools to monitor time spent on the app.
- iv) Ethical Advertising Practices:** Advertisements should not manipulate users but rather provide genuine value, ensuring that users are not exploited for profit.

In conclusion, Sudo's current revenue model raises significant ethical concerns. Implementing an ethical policy that prioritizes user transparency and well-being is crucial for fostering trust and accountability.

5. Mr. Ross Geller is a Final year Undergraduate student doing his thesis in the domain of Natural Language Processing and Social Media Analytics. His topic is "Identify Cyberbullying and Offensive posts from Social Networking Sites (SNS)". He wants to collect Social Media data of 10,000 Facebook users and utilize it to train the model. To collect the Data, He needs to prepare a consent form for the participants mentioning the necessary information of the research and assuring the anonymity of their data along with maintaining few other ethical issues.

Your task is to prepare the consent form for Mr. Ross to start the data collection process. (The Form must not be more than two-pages long)

Ans:

Consent Form for Data Collection

Title of Research: Identifying Cyberbullying and Offensive Posts on Social Networking Sites (SNS)

Researcher: Mr. Ross Geller

Contact Information: [Email Address]

Institution: [University Name]

Introduction: You are being invited to participate in a research study aimed at identifying cyberbullying and offensive posts on social media. This study is part of Mr. Ross Geller's undergraduate thesis.

Purpose of the Study: The goal of this research is to collect and analyze social media data from Facebook users to develop a model that can effectively identify harmful online behavior.

Data Collection:

- Your participation involves sharing your social media activity and posts.
- We will collect data from 10,000 Facebook users.
- Your responses will be anonymized, meaning your personal information will not be linked to the data collected.

Voluntary Participation:

- Participation in this study is entirely voluntary.
- You may withdraw your consent at any time without any consequences.

Confidentiality:

- All data collected will be kept confidential and secure.
- Only Mr. Geller and his research team will have access to the data.

Risks and Benefits:

- While there are minimal risks associated with this study, some participants may feel uncomfortable discussing their social media activities.
- The findings of this research may contribute to better understanding and addressing cyberbullying on social media platforms.

Consent: By signing below, you indicate that you understand the purpose of the study, the nature of your participation, and your rights as a participant. You consent to participate in this research study.

Participant Signature: _____

Date: _____

Researcher Signature: _____

Date: _____

Note: For any questions regarding this research, please feel free to contact Mr. Ross Geller at [Email Address].

Thank you for your consideration!